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Adaptive Tx-Rx crosstalk cancellation for radar systems

Abstract

In an embodiment, a method includes: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4241347	12/1979	Albanese et al.	N/A	N/A
6147572	12/1999	Kaminski et al.	N/A	N/A
6414631	12/2001	Fujimoto	N/A	N/A
6636174	12/2002	Arikan et al.	N/A	N/A
7048973	12/2005	Sakamoto et al.	N/A	N/A
7057564	12/2005	Tsai et al.	N/A	N/A
7171052	12/2006	Park	N/A	N/A
7317417	12/2007	Arikan et al.	N/A	N/A
7596241	12/2008	Rittscher et al.	N/A	N/A
7692574	12/2009	Nakagawa	N/A	N/A
7873326	12/2010	Sadr	N/A	N/A
7889147	12/2010	Tam et al.	N/A	N/A
8228382	12/2011	Pattikonda	N/A	N/A
8497805	12/2012	Rofougaran et al.	N/A	N/A
8659369	12/2013	Rofougaran et al.	N/A	N/A
8731502	12/2013	Salle et al.	N/A	N/A
8836596	12/2013	Richards et al.	N/A	N/A
8847814	12/2013	Himmelstoss et al.	N/A	N/A
8860532	12/2013	Gong et al.	N/A	N/A
8976061	12/2014	Chowdhury	N/A	N/A
9172132	12/2014	Kam et al.	N/A	N/A
9182476	12/2014	Wintermantel	N/A	N/A
9202105	12/2014	Wang et al.	N/A	N/A
9229102	12/2015	Wright et al.	N/A	N/A
9413079	12/2015	Kamgaing et al.	N/A	N/A
9495600	12/2015	Heu et al.	N/A	N/A
9886095	12/2017	Pothier	N/A	N/A
9935065	12/2017	Baheti et al.	N/A	N/A
10795012	12/2019	Santra et al.	N/A	N/A
2003/0179127	12/2002	Wienand	N/A	N/A
2004/0238857	12/2003	Beroz et al.	N/A	N/A
2006/0001572	12/2005	Gaucher et al.	N/A	N/A
2006/0049995	12/2005	Imaoka et al.	N/A	N/A
2006/0067456	12/2005	Ku et al.	N/A	N/A
2007/0028219	12/2006	Miller	717/124	G05B 23/0275
2007/0210959	12/2006	Herd et al.	N/A	N/A

2008/0106460	12/2007	Kurtz et al.	N/A	N/A
2008/0238759	12/2007	Carocari et al.	N/A	N/A
2008/0291115	12/2007	Doan et al.	N/A	N/A
2008/0308917	12/2007	Pressel et al.	N/A	N/A
2009/0073026	12/2008	Nakagawa	N/A	N/A
2009/0085815	12/2008	Jakab et al.	N/A	N/A
2009/0146865	12/2008	Watanabe et al.	N/A	N/A
2009/0153428	12/2008	Rofougaran et al.	N/A	N/A
2009/0262005	12/2008	McNeill et al.	N/A	N/A
2009/0315761	12/2008	Walter et al.	N/A	N/A
2010/0207805	12/2009	Haworth	N/A	N/A
2011/0299433	12/2010	Darabi et al.	N/A	N/A
2012/0087230	12/2011	Guo et al.	N/A	N/A
2012/0092284	12/2011	Rofougaran et al.	N/A	N/A
2012/0116231	12/2011	Liao et al.	N/A	N/A
2012/0195161	12/2011	Little et al.	N/A	N/A
2012/0206339	12/2011	Dahl	N/A	N/A
2012/0265486	12/2011	Klofer et al.	N/A	N/A
2012/0268314	12/2011	Kuwahara et al.	N/A	N/A
2012/0280900	12/2011	Wang et al.	N/A	N/A
2013/0027240	12/2012	Chowdhury	N/A	N/A
2013/0106673	12/2012	McCormack et al.	N/A	N/A
2014/0028542	12/2013	Lovitt et al.	N/A	N/A
2014/0070994	12/2013	Schmalenberg et al.	N/A	N/A
2014/0145883	12/2013	Baks et al.	N/A	N/A
2014/0324888	12/2013	Xie et al.	N/A	N/A
2015/0181840	12/2014	Tupin, Jr. et al.	N/A	N/A
2015/0185316	12/2014	Rao et al.	N/A	N/A
2015/0212198	12/2014	Nishio et al.	N/A	N/A
2015/0243575	12/2014	Strothmann et al.	N/A	N/A
2015/0277569	12/2014	Sprenger et al.	N/A	N/A
2015/0325925	12/2014	Kamgaing et al.	N/A	N/A
2015/0346820	12/2014	Poupyrev et al.	N/A	N/A
2015/0348821	12/2014	Wanaga et al.	N/A	N/A
2015/0364816	12/2014	Murugan et al.	N/A	N/A
2016/0018511	12/2015	Nayyar et al.	N/A	N/A
2016/0041617	12/2015	Poupyrev	N/A	N/A
2016/0041618	12/2015	Poupyrev	N/A	N/A
2016/0061942	12/2015	Rao et al.	N/A	N/A
2016/0061947	12/2015	Patole et al.	N/A	N/A
2016/0098089	12/2015	Poupyrev	N/A	N/A
2016/0103213	12/2015	Kram et al.	N/A	N/A
2016/0109566	12/2015	Liu et al.	N/A	N/A
2016/0118353	12/2015	Ahrens et al.	N/A	N/A
2016/0135655	12/2015	Ahn et al.	N/A	N/A
2016/0146931	12/2015	Rao et al.	N/A	N/A
2016/0146933	12/2015	Rao et al.	N/A	N/A
2016/0178730	12/2015	Trotta et al.	N/A	N/A
2016/0187462	12/2015	Altus et al.	N/A	N/A
2016/0191232	12/2015	Subburaj et al.	N/A	N/A

2016/0223651	12/2015	Kamo et al.	N/A	N/A
2016/0240907	12/2015	Haroun	N/A	N/A
2016/0249133	12/2015	Sorensen	N/A	N/A
2016/0252607	12/2015	Saboo et al.	N/A	N/A
2016/0259037	12/2015	Molchanov et al.	N/A	N/A
2016/0266233	12/2015	Mansour	N/A	N/A
2016/0269815	12/2015	Liao et al.	N/A	N/A
2016/0291130	12/2015	Ginsburg et al.	N/A	N/A
2016/0299215	12/2015	Dandu et al.	N/A	N/A
2016/0306034	12/2015	Trotta et al.	N/A	N/A
2016/0320852	12/2015	Poupyrev	N/A	N/A
2016/0320853	12/2015	Lien et al.	N/A	N/A
2016/0327633	12/2015	Kumar Y.B. et al.	N/A	N/A
2016/0334502	12/2015	Ali et al.	N/A	N/A
2016/0349845	12/2015	Poupyrev et al.	N/A	N/A
2017/0033062	12/2016	Liu et al.	N/A	N/A
2017/0045607	12/2016	Bharadwaj et al.	N/A	N/A
2017/0052618	12/2016	Lee et al.	N/A	N/A
2017/0054449	12/2016	Mani et al.	N/A	N/A
2017/0060254	12/2016	Molchanov et al.	N/A	N/A
2017/0070952	12/2016	Balakrishnan et al.	N/A	N/A
2017/0074974	12/2016	Rao et al.	N/A	N/A
2017/0074980	12/2016	Adib et al.	N/A	N/A
2017/0090014	12/2016	Subburaj et al.	N/A	N/A
2017/0090015	12/2016	Breen et al.	N/A	N/A
2017/0115377	12/2016	Giannini et al.	N/A	N/A
2017/0131395	12/2016	Reynolds et al.	N/A	N/A
2017/0139036	12/2016	Nayyar et al.	N/A	N/A
2017/0141453	12/2016	Waelde et al.	N/A	N/A
2017/0170947	12/2016	Yang	N/A	N/A
2017/0176574	12/2016	Eswaran et al.	N/A	N/A
2017/0192847	12/2016	Rao et al.	N/A	N/A
2017/0201019	12/2016	Trotta	N/A	N/A
2017/0212597	12/2016	Mishra	N/A	N/A
2017/0364160	12/2016	Malysa et al.	N/A	N/A
2018/0046255	12/2017	Rothera et al.	N/A	N/A
2018/0071473	12/2017	Trotta et al.	N/A	N/A
2018/0101239	12/2017	Yin et al.	N/A	N/A
2020/0191911	12/2019	Meissner	N/A	G01S 13/343
2020/0217926	12/2019	Pietsch et al.	N/A	N/A
2020/0264272	12/2019	Visweswaran	N/A	H01Q 1/525
2022/0350010	12/2021	Sagi	N/A	G01S 13/04
2023/0139751	12/2022	Sanderovich	382/103	G01S 13/89

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1463161	12/2002	CN	N/A
1716695	12/2005	CN	N/A
101490578	12/2008	CN	N/A
101585361	12/2008	CN	N/A

102788969	12/2011	CN	N/A
102967854	12/2012	CN	N/A
103529444	12/2013	CN	N/A
203950036	12/2013	CN	N/A
102008054570	12/2009	DE	N/A
102011100907	12/2011	DE	N/A
102011075725	12/2011	DE	N/A
102014118063	12/2014	DE	N/A
2247799	12/1991	GB	N/A
2001174539	12/2000	JP	N/A
2004198312	12/2003	JP	N/A
2006234513	12/2005	JP	N/A
2008029025	12/2007	JP	N/A
2008089614	12/2007	JP	N/A
2009069124	12/2008	JP	N/A
2011529181	12/2010	JP	N/A
2012112861	12/2011	JP	N/A
2013521508	12/2012	JP	N/A
2014055957	12/2013	JP	N/A
20090063166	12/2008	KR	N/A
20140082815	12/2013	KR	N/A
2007060069	12/2006	WO	N/A
2013009473	12/2012	WO	N/A
2016033361	12/2015	WO	N/A

OTHER PUBLICATIONS

Will, Christoph et al., “Advanced Template Matching Algorithm for Instantaneous Heartbeat Detection using Continuous Wave Radar Systems”, ResearchGate, May 2017, 5 pages. cited by applicant

Will, Christoph et al., “Human Target Detection, Tracking, and Classification Using 24-GHz FMCW Radar”, IEEE Sensors Journal, vol. 19, No. 17, Sep. 1, 2019, pp. 7283-7299. cited by applicant

Will, Christoph et al., “Local Pulse Wave Detection using Continuous Wave Radar Systems”, IEEE Journal of Electromagnetics, RF and Microwaves in Medicine and Biology, Oct. 25, 2017, 9 pages. cited by applicant

Will, Christoph et al., “Radar-Based Heart Sound Detection”, Scientific Reports, www.nature.com/scientificreports, Jul. 26, 2018, 15 pages. cited by applicant

Xin, Qin et al., “Signal Processing for Digital Beamforming FMCW SAR,” Hindawi Publishing Corporation, Mathematical Problems in Engineering, vol. 2014, Article ID 859890, <http://dx.doi.org/10.1155/2014/859890>, Apr. 15, 2014, 11 pages. cited by applicant

“BT24MTR11 Using BGT24MTR11 in Low Power Applications 24 GHz Rader,” Application Note AN341, Revision: Rev 1.0, Infineon Technologies AG, Munich, Germany, Dec. 2, 2013, 25 pages. cited by applicant

Chen, Xiaolong et al., “Detection and Extraction of Marine Target with Micromotion via Short-Time Fractional Fourier Transform in Sparse Domain,” IEEE International Conference on Signal Processing, Communications and Computing, ICSPCC, Aug. 5-8, 2016, 5 pages. cited by applicant

Chen, Xiaolong et al., “Detection and Extraction of Target with Micromotion in Spiky Sea Clutter via Short-Time Fractional Fourier Transform”, IEEE Transactions on Geoscience and Remote Sensing, vol. 52, No. 2, Feb. 2014, pp. 1002-1018. cited by applicant

Chioukh, Lydia et al., “Noise and Sensitivity of Harmonic Radar Architecture for Remote Sensing

and Detection of Vital Signs”, IEEE Transactions on Microwave Theory and Techniques, vol. 62, No. 9, Sep. 2014, pp. 1847-1855. cited by applicant

Chuanhua, Du, “FMCW Radar Range-Doppler Processing and Beam Formation Technology,” Chinese Doctoral Dissertations & Master's Theses Full Text Database (Masters)—Information Science and Technology Series, China National Knowledge Infrastructure, ISSN 1674-0246, CN 11-9144/G, Dec. 16, 2004-Mar. 2015, 14 pages. cited by applicant

Deacon, Peter et al., “Frequency Modulated Continuous Wave (FMCW) Radar,” Design Team 6 Technical Lecture, Nov. 9, 2011, 27 pages. cited by applicant

Dham, Vivek “Programming Chirp Parameters in TI Radar Devices,” Application Report SWRA553, Texas Instruments, May 2017, 15 pages. cited by applicant

Diederichs, Kailtyn et al., “Wireless Biometric Individual Identification Utilizing Millimeter Waves”, IEEE Sensors Letters, vol. 1, No. 1, IEEE Sensors Council 3500104, Feb. 2017, 4 pages. cited by applicant

Gigie, Andrew et al., “Novel Approach for Vibration Detection Using Indented Radar”, Progress in Electromagnetic Research C, vol. 87, pp. 147-162, Oct. 4, 2018. cited by applicant

Gouveia, Carolina et al., “A Review on Methods for Random Motion Detection and Compensation in Bio-Radar Systems”, Sensors, MDPI, Jan. 31, 2019, 17 pages. cited by applicant

Gu, Changzhan et al., “Assessment of Human Respiration Patterns via Noncontact Sensing Using Doppler Multi-Radar System”, Sensors Mar. 2015, 15(3), 6383-6398, doi: 10.3390/s150306383, 17 pages. cited by applicant

Gu, Changzhan et al., “Deep Neural Network based Body Movement Cancellation for Doppler Radar Vital Sign Detection”, IEEE MTT-S International Wireless Symposium (IWS) May 19-22, 2019, 3 pages. cited by applicant

Gu, Changzhu “Short-Range Noncontact Sensors for Healthcare and Other Emerging Applications: A Review”, Sensors, MDPI, Jul. 26, 2016, 24 pages. cited by applicant

Gu, Changzhan et al., “From Tumor Targeting to Speed Monitoring”, IEEE Microwave Magazine, ResearchGate, Jun. 2014, 11 pages. cited by applicant

Guercan, Yalin “Super-resolution Algorithms for Joint Range-Azimuth-Doppler Estimation in Automotive Radars,” Technische Universitet Delft, TUDelft University of Technology Challenge the Future, Jan. 25, 2017, 72 pages. cited by applicant

Hu, Wei et al., “Noncontact Accurate Measurement of Cardiopulmonary Activity Using a Compact Quadrature Doppler Radar Sensor”, IEEE Transactions on Biomedical Engineering, vol. 61, No. 3, Mar. 2014, pp. 725-735. cited by applicant

Immooeev, I. Ya. “Ultrawideband Radars: Features and Capabilities”, Journal of Communications Technology and Electronics, ISSN: 1064-2269, vol. 54, No. 1, Feb. 8, 2009, pp. 1-26. cited by applicant

Inac, Ozgur et al., “A Phased Array RFIC with Built-In Self-Test Capabilities,” IEEE Transactions on Microwave Theory and Techniques, vol. 60, No. 1, Jan. 2012, 10 pages. cited by applicant

Killedar, Abdulraheem “XWR1xxx Power Management Optimizations—Low Cost LC Filter Solution,” Application Report SWRA577, Texas Instruments, Oct. 2017, 19 pages. cited by applicant

Kishore, N. et al., “Millimeter Wave Antenna for Intelligent Transportation Systems Application”, Journal of Microwaves, Optoelectronics and Electromagnetic Applications, vol. 17, No. 1, Mar. 2018, pp. 171-178. cited by applicant

Kizhakkal, V., “Pulsed Radar Target Recognition Based on Micro-Doppler Signatures Using Wavelet Analysis”, A Thesis, Graduate Program in Electrical and Computer Engineering, Ohio State University, Jan. 2013-May 2013, 118 pages. cited by applicant

Kuehnke, Lutz, “Phased Array Calibration Procedures Based on Measured Element Patterns,” 2001 Eleventh International Conference on Antennas and Propagation, IEEE Conf., Publ. No. 480, Apr. 17-20, 2001, 4 pages. cited by applicant

Li, Changzhi et al., "A Review on Recent Advances in Doppler Radar Sensors for Noncontact Healthcare Monitoring", IEEE Transactions on Microwave Theory and Techniques, vol. 61, No. 5, May 2013, pp. 2046-2060. cited by applicant

Li, Changzhi et al., "A Review on Recent Progress of Portable Short-Range Noncontact Microwave Radar Systems", IEEE Transactions on Microwave Theory and Techniques, vol. 65, No. 5, May 2017, pp. 1692-1706. cited by applicant

Li, Changzhi et al., "Random Body Movement Cancellation in Doppler Radar Vital Sign Detection", IEEE Transactions on Microwave Theory and Techniques, vol. 56, No. 12, Dec. 2008, pp. 3143-3152. cited by applicant

Li, Changzhi et al., "Robust Overnight Monitoring of Human Vital Signs by a Non-contact Respiration and Heartbeat Detector", IEEE Proceedings of the 28th EMBS Annual International Conference, FrA05.5, Aug. 30-Sep. 3, 2006, 4 pages. cited by applicant

Li, Changzhi "Vital-sign monitoring on the go", Sensors news and views, www.nature.com/natureelectronics, Nature Electronics, vol. 2, Jun. 2019, 2 pages. cited by applicant

Lim, Soo-Chul et al., "Expansion of Smartwatch Touch Interface from Touchscreen to Around Device Interface Using Infrared Line Image Sensors," Sensors 2015, ISSN 1424-8220, vol. 15, 16642-16653, doi:10.3390/s150716642, www.mdpi.com/journal/sensors, Jul. 15, 2009, 12 pages. cited by applicant

Lin, Jau-Jr et al., "Design of an FMCW radar baseband signal processing system for automotive application," SpringerPlus a SpringerOpen Journal, (2016) 5:42, <http://creativecommons.org/licenses/by/4.0/>, DOI 10.1186/\$40064-015-1583-5; Jan. 2016, 16 pages. cited by applicant

Massagram, Wansuree et al., "Assessment of Heart Rate Variability and Respiratory Sinus Arrhythmia via Doppler Radar", IEEE Transactions on Microwave Theory and Techniques, vol. 57, No. 10, Oct. 2009, pp. 2542-2549. cited by applicant

Mercuri, Marco et al., "Vital-sign monitoring and spatial tracking of multiple people using a contactless radar-based sensor", Nature Electronics, vol. 2, Articles, <https://doi.org/10.1038/s41928-019-0258-6>, Jun. 2019, 13 pages. cited by applicant

Microwave Journal Frequency Matters, "Single-Chip 24 GHz Radar Front End," Infineon Technologies AG, www.microwavejournal.com/articles/print/21553-single-chip-24-ghz-radar-front-end, Feb. 13, 2014, 2 pages. cited by applicant

Mostov, K., et al., "Medical applications of shortwave FM radar: Remote monitoring of cardiac and respiratory motion", Am. Assoc. Phys. Med., 37(3), Mar. 2010, pp. 1332-1338. cited by applicant

Oguntala, G et al., "Indoor location identification technologies for real-time IoT-based applications: an inclusive survey", Elsevier Inc., <http://hdl.handle.net/10454/16634>, Oct. 2018, 21 pages. cited by applicant

Peng, Zhengyu et al., "A Portable FMCW Interferometry Radar with Programmable Low-IF Architecture for Localization, ISAR Imaging, and Vial Sign Tracking", IEEE Transactions on Microwave Theory and Techniques, Dec. 15, 2016, 11 pages. cited by applicant

Qadir, Shahida G., et al., "Focused ISAR Imaging of Rotating Target in Far-Field Compact Range Anechoic Chamber," 14th International Conference on Aerospace Sciences & Aviation Technology, ASAT-14-241-IP, May 24-26, 2011, 7 pages. cited by applicant

Richards, Mark A., "Fundamentals of Radar Signal Processing," McGraw Hill Electronic Engineering, ISBN: 0-07-144474-2, Jun. 2005, 93 pages. cited by applicant

Sakamoto, Takuya et al., "Feature-Based Correlation and Topological Similarity for Interbeat Interval Estimation Using Ultrawideband Radar", IEEE Transactions on Biomedical Engineering, vol. 63, No. 4, Apr. 2016, pp. 747-757. cited by applicant

Santra, Avik et al., "Short-range multi-mode continuous-wave radar for vital sign measurement and imaging", ResearchGate, Conference Paper, Apr. 2018, 6 pages. cited by applicant

Schroff, Florian et al., "FaceNet: A Unified Embedding for Face Recognition and Clustering,"

CVF, CVPR2015, IEEE Computer Society Conference on Computer Vision and Pattern Recognition; Mar. 12, 2015, pp. 815-823. cited by applicant

Simon, W., et al., "Highly Integrated KA-Band Tx Frontend Module Including 8x8 Antenna Array," IMST GmbH, Germany, Asia Pacific Microwave Conference, Dec. 7-10, 2009, 63 pages. cited by applicant

Singh, Aditya et al., "Data-Based Quadrature Imbalance Compensation for a CW Doppler Radar System", <https://www.researchgate.net/publication/258793573>, IEEE Transactions on Microwave Theory and Techniques, Apr. 2013, 7 pages. cited by applicant

Suleymanov, Suleyman, "Design and Implementation of an FMCW Radar Signal Processing Module for Automotive Applications," Master Thesis, University of Twente, Aug. 31, 2016, 64 pages. cited by applicant

Thayaparan, T. et al., "Micro-Doppler Radar Signatures for Intelligent Target Recognition," Defence Research and Development Canada, Technical Memorandum, DRDC Ottawa TM 2004-170, Sep. 2004, 73 pages. cited by applicant

Thayaparan, T et al., "Intelligent target recognition using micro-Doppler radar signatures," Defence R&D Canada, Radar Sensor Technology III, Proc. of SPIE, vol. 7308, 730817, Dec. 9, 2009, 11 pages. cited by applicant

Tu, Jianxuan et al., "Fast Acquisition of Heart Rate in Noncontact Vital Sign Radar Measurement Using Time-Window-Variation Technique", IEEE Transactions on Instrumentation and Measurement, vol. 65, No. 1, Jan. 2016, pp. 112-122. cited by applicant

Vinci, Gabor et al., "Microwave Interferometer Radar-Based Vital Sign Detection for Driver Monitoring Systems", IEEE MTT-S International Conference on Microwaves for Intelligent Mobility, Apr. 27-29, 2015, 4 pages. cited by applicant

Vinci, Gabor et al., "Six-Port Radar Sensor for Remote Respiration Rate and Heartbeat Vital-Sign Monitoring", IEEE Transactions on Microwave Theory and Techniques, vol. 61, No. 5, May 2013, pp. 2093-2100. cited by applicant

Wang, Fu-Kang et al., "Wrist Pulse Rate Monitor Using Self-Injection-Locked Radar Technology", Biosensors, MDPI, Oct. 26, 2016, 12 pages. cited by applicant

Wilder, Carol N., et al., "Respiratory patterns in infant cry," Canada Journal of Speech, Human Communication Winter, 1974-75, http://cjslpa.ca/files/1974_HumComm_Vol_01/No_03_2-60/Wilder_Baken_HumComm_1974.pdf, 1974, pp. 18-34. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to an electronic system and method, and, in particular embodiments, to adaptive transmitter-receiver crosstalk cancellation for radar systems.

BACKGROUND

(2) Applications in the millimeter-wave frequency regime have gained significant interest in the past few years due to the rapid advancement in low-cost semiconductor technologies, such as silicon germanium (SiGe) and fine geometry complementary metal-oxide semiconductor (CMOS) processes. Availability of high-speed bipolar and metal-oxide semiconductor (MOS) transistors has

led to a growing demand for integrated circuits for millimeter-wave applications at e.g., 24 GHz, 60 GHz, 77 GHz, and 80 GHz and also beyond 100 GHz. Such applications include, for example, automotive radar systems and multi-gigabit communication systems.

(3) In some radar systems, the distance between the radar and a target is determined by transmitting a frequency modulated signal, receiving a reflection of the frequency modulated signal (also referred to as the echo), and determining a distance based on a time delay and/or frequency difference between the transmission and reception of the frequency modulated signal. Accordingly, some radar systems include a transmit antenna to transmit the radio-frequency (RF) signal, and a receive antenna to receive the reflected RF signal, as well as the associated RF circuits used to generate the transmitted signal and to receive the RF signal. In some cases, multiple antennas may be used to implement directional beams using phased array techniques. A multiple-input and multiple-output (MIMO) configuration with multiple chipsets can be used to perform coherent and non-coherent signal processing as well.

(4) Crosstalk from the transmit path of a radar to the receive path of the radar may occur, e.g., when a portion of the transmitted signal is received by the receive path via an internal path of the radar. Such crosstalk is commonly referred to as TX-RX crosstalk or TX-RX leakage. A standard approach to cope with this crosstalk is to use high-pass filters (HPFs). Another approach uses a moving target indication (MTI) in applications with moving targets.

SUMMARY

(5) In accordance with an embodiment, a method includes: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

(6) In accordance with an embodiment, a radar system includes: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and a processing system configured to: initialize a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, select a set of data from the radar data, perform target detection on the set of data, and after performing the target detection on the set of data, when no target is detected in the set of data, update the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generate a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generate the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

(7) In accordance with an embodiment, a method includes: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the a transmit path of a millimeter-wave radar sensor and a receive path of the millimeter-wave radar sensor; receiving radar data from the millimeter-wave radar sensor; selecting a sub-set of data from the radar data; determining a polynomial fit on the sub-set of data; determining a residual of the polynomial fit based on the polynomial fit and the sub-set of data; and determining a spectrum of the residual of the polynomial fit; incoherently integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than a maximum number of bins of the spectrum of the residual of the polynomial fit; performing target detection on the sub-set of data based on the integrated spectrum; after performing the target detection on the sub-set of data, when no target is detected in the sub-set of data, updating the

crosstalk compensation factor based on the sub-set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the sub-set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; and performing target detection or target tracking based on the radar spectrum.

(8) In accordance with an embodiment, a method includes: receiving radar data from a radar sensor; selecting a set of data from the radar data; determining a crosstalk approximation to a transmitter-receiver crosstalk between a transmit path of the radar sensor and a receive path of the radar sensor based on the set of data; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; determining a spectrum of the residual of the crosstalk approximation; and performing target detection on the set of data based on the spectrum of the residual of the crosstalk approximation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 shows a schematic diagram of a millimeter-wave radar system, according to an embodiment of the present invention;

(3) FIG. 2 shows a sequence of chirps transmitted by the radar system of FIG. 1, according to an embodiment of the present invention;

(4) FIGS. 3 and 4 show exemplary range spectra associated with the millimeter-wave radar sensor of FIG. 1, according to embodiments of the present invention;

(5) FIG. 5 shows waveforms of radar data, according to an embodiment of the present invention; and

(6) FIGS. 6 and 7 show flow charts of embodiment methods for dynamic crosstalk cancellation, according to an embodiment of the present invention.

(7) Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(8) The making and using of the embodiments disclosed are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

(9) The description below illustrates the various specific details to provide an in-depth understanding of several example embodiments according to the description. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials and the like. In other cases, known structures, materials or operations are not shown or described in detail so as not to obscure the different aspects of the embodiments. References to “an embodiment” in this description indicate that a particular configuration, structure or feature described in relation to the embodiment is included in at least one embodiment. Consequently, phrases such as “in one embodiment” that may appear at different points of the present description do not necessarily refer exactly to the same embodiment. Furthermore, specific formations, structures or features may be combined in any appropriate manner in one or more embodiments.

(10) Embodiments of the present invention will be described in specific contexts, e.g., an adaptive TX-RX crosstalk cancellation for millimeter-wave radars operated as a frequency-modulated continuous wave (FMCW) radar. Embodiments of the present invention may be used with other types of radars systems, such as radars operating outside the millimeter-wave range.

(11) In an embodiment of the present invention, a millimeter-wave radar adaptively cancels the effect of crosstalk from the transmit path to the receive path on a radar spectrum based on whether close-range targets are detected. A set of radar data is selected from a set of radar data. If no close-range targets are detected in the set of radar data, a crosstalk compensation factor is updated. If a close-range target is detected, the crosstalk compensation factor is not updated. The radar spectrum is compensated by cancelling the contribution of the crosstalk (e.g., in the time domain or in the frequency domain) based on the crosstalk compensation factor.

(12) FIG. 1 shows a schematic diagram of millimeter-wave radar system **100**, according to an embodiment of the present invention. Millimeter-wave radar system **100** includes millimeter-wave radar sensor **102** and processing system **104**. In some embodiments, millimeter-wave radar sensor **102** may be used to generate radar data ($x_{\text{sub.out_dig}}(n)$) and processing system **104** may implement a radar processing chain, e.g., to detect, track, and/or classify targets.

(13) During normal operation, millimeter-wave radar sensor **102** operates as a frequency-modulated continuous-wave (FMCW) radar sensor and transmits a plurality of radar signals **106**, such as chirps, towards scene **120** using one or more transmitter (TX) antennas **114**. The radar signals **106** are generated using RF and analog circuits **130**. The radar signals **106** may be, e.g., in the 20 GHz to 122 GHz range. Other frequencies may also be used.

(14) The objects in scene **120** may include one or more static or moving objects, such as cars, motorcycles, bicycles, trucks, and other vehicles, idle and moving humans and animals, poles, furniture, machinery, mechanical structures, walls and other types of structures. Other objects may also be present in scene **120**.

(15) The radar signals **106** are reflected by objects in scene **120**. The reflected radar signals **108**, which are also referred to as the echo signal, are received by one or more receiving (RX) antennas **116**. RF and analog circuits **130** processes the received reflected radar signals **108** using, e.g., band-pass filters (BPFs), low-pass filters (LPFs), mixers, low-noise amplifier (LNA), and/or intermediate frequency (IF) amplifiers in ways known in the art to generate an analog signal $x_{\text{sub.out}}(t)$.

(16) The analog signal $x_{\text{sub.out}}(t)$ is converted to raw digital data $x_{\text{sub.out_dig}}(n)$ (also referred to as radar data) using ADC **112**. The raw digital data $x_{\text{sub.out_dig}}(n)$ is processed by processing system **104** to, e.g., detect, track, and/or classify targets.

(17) Controller **no** controls one or more circuits of millimeter-wave radar sensor **102**, such as RF and analog circuit **130** and/or ADC **112**. Controller **no** may be implemented, e.g., as a custom digital or mixed signal circuit, for example. Controller **no** may also be implemented in other ways, such as using a custom or general purpose processor or controller configured to execute instructions stored in a memory. In some embodiments, processing system **104** implements a portion or all of controller **no**.

(18) Processing system **104** may be implemented with a general purpose processor, controller or digital signal processor (DSP) that includes, for example, combinatorial circuits coupled to a memory. In some embodiments, processing system **104** may be implemented as an application specific integrated circuit (ASIC). In some embodiments, processing system **104** may be implemented with an ARM, RISC, or x86 architecture, for example. In some embodiments, processing system **104** may include an artificial intelligence (AI) accelerator. Some embodiments may use a combination of hardware accelerator and software running on a DSP or general purpose microcontroller. In some embodiments, processing system **104** may be implemented with a plurality of processors and/or controllers. Other implementations are also possible.

(19) In some embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented inside the same integrated circuit (IC). For example, in some

embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in respective semiconductor substrates that are integrated in the same package. In other embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in the same monolithic semiconductor substrate. In some embodiments, millimeter-wave radar sensor **102** and processing system **104** are implemented in respective integrated circuits. In some embodiments, a plurality of integrated circuits is used to implement millimeter-wave radar sensor **102**. In some embodiments, a plurality of integrated circuits is used to implement processing system **104**. Other implementations are also possible.

(20) As a non-limiting example, RF and analog circuits **130** may be implemented, e.g., as shown in FIG. **1**. During normal operation, voltage-controlled oscillator (VCO) **136** generates radar signals, such as a linear frequency chirps (e.g., from 57 GHz to 64 GHz, or from 76 GHz to 77 GHz), which are transmitted by the transmitting antenna **114**. The VCO **136** is controlled by PLL **134**, which receives a reference clock signal (e.g., 80 MHz) from reference oscillator **132**. PLL **134** is controlled by a loop that includes frequency divider **138** and amplifier **140**. Amplifiers **137** may be used to drive transmitting antenna **114**.

(21) The TX radar signals **106** transmitted by transmitting antenna **114** are reflected by objects in scene **120** and received by receiving antenna **116**. The echo received by receiving antenna **116** is mixed with a replica of the signal transmitted by transmitting antenna **114** using mixer **146** to produce intermediate frequency (IF) signal $x_{\text{sub}}.\text{IF}(t)$ (also known as beat signal). In some embodiments, the beat signal $x_{\text{sub}}.\text{IF}(t)$ has a bandwidth between 10 kHz and 1 MHz. Beat signals with a bandwidth lower than 10 kHz or higher than 1 MHz is also possible. Amplifier **145** may be used to receive the reflected radar signals from antenna **116**.

(22) Beat signal $x_{\text{sub}}.\text{IF}(t)$ is filtered with low-pass filters (LPF) **148** and then sampled by ADC **112**. ADC **112** is advantageously capable of sampling the filtered beat signal $x_{\text{sub}}.\text{out}(t)$ with a sampling frequency that is much smaller than the frequency of the signal received by receiving antenna **116**. Using FMCW radars, therefore, advantageously allows for a compact and low cost implementation of ADC **112**, in some embodiments.

(23) The raw digital data $x_{\text{sub}}.\text{out_dig}(n)$, which in some embodiments include the digitized version of the filtered beat signal $x_{\text{sub}}.\text{out}(t)$ is (e.g., temporarily) stored, e.g., in matrices of $N_{\text{sub}.c} \times N_{\text{sub}.s}$ per receiving antenna **116**, where $N_{\text{sub}.c}$ is the number of chirps considered in a frame and $N_{\text{sub}.s}$ is the number of samples per chirp, for further processing by processing system **104**.

(24) FIG. **2** shows a sequence of chirps **106** transmitted by TX antenna **114**, according to an embodiment of the present invention. As shown by FIG. **2**, chirps **106** are organized in a plurality of frames and may be implemented as up-chirps. Some embodiments may use down-chirps or a combination of up-chirps and down-chirps, such as up-down chirps and down-up chirps. Other waveform shapes may also be used.

(25) As shown in FIG. **2**, each frame may include a plurality of chirps **106**. For example, in some embodiments, the number of chirps in a frame is 16. Some embodiments may include more than 16 chirps per frame, such as 20 chirps, 32 chirps, or more, or less than 16 chirps per frame, such as 10 chirps, 8 chirps, or less. In some embodiments, each frame includes only a single chirp.

(26) In some embodiments, frames are repeated every FT time. In some embodiments, FT time is 50 ms. A different FT time may also be used, such as more than 50 ms, such as 60 ms, 100 ms, 200 ms, or more, or less than 50 ms, such as 45 ms, 40 ms, or less.

(27) In some embodiments, the FT time is selected such that the time between the beginning of the last chirp of frame n and the beginning of the first chirp of frame $n+1$ is equal to PRT. Other embodiments may use or result in a different timing.

(28) The time between chirps of a frame is generally referred to as pulse repetition time (PRT). In some embodiments, the PRT is 5 ms. A different PRT may also be used, such as less than 5 ms, such as 4 ms, 2 ms, or less, or more than 5 ms, such as 6 ms, or more.

(29) The duration of the chirp (from start to finish) is generally referred to as chirp time (CT). In some embodiments, the chirp time may be, e.g., 64 μ s. Higher chirp times, such as 128 μ s, or higher, may also be used. Lower chirp times, may also be used.

(30) In some embodiments, the chirp bandwidth may be, e.g., 4 GHz. Higher bandwidth, such as 6 GHz or higher, or lower bandwidth, such as 2 GHz, 1 GHz, or lower, may also be possible.

(31) In some embodiments, the sampling frequency of millimeter-wave radar sensor **902** may be, e.g., 1 MHz. Higher sampling frequencies, such as 2 MHz or higher, or lower sampling frequencies, such as 500 kHz or lower, may also be possible.

(32) In some embodiments, the number of samples used to generate a chirp may be, e.g., 64 samples. A higher number of samples, such as 128 samples, or higher, or a lower number of samples, such as 32 samples or lower, may also be used.

(33) As signal transmission and reception is performed simultaneously, crosstalk may occur between transmit path **113** and receive path **115**. Crosstalk from transmit path **113** to receive path **115** generally covers the lower part of the range spectrum with high amplitudes. For example, FIG. **3** shows exemplary range spectrum **300** associated with millimeter-wave radar sensor **102**, according to an embodiment of the present invention. Range spectrum **300** is obtained from a fast Fourier transform (FFT) and non-coherent integration of all chirps and all channels of the radar data ($x_{\text{sub.out_dig}}(n)$) of a single frame with $N_{\text{sub.S}}=128$, $N_{\text{sub.c}}=16$, $N_{\text{sub.R}}=8$ and $BW=4$ GHz, where $N_{\text{sub.c}}$ is the number of chirps considered in a frame, $N_{\text{sub.s}}$ is the number of samples per chirp, and $N_{\text{sub.R}}$ is the number of receiving antennas **116**, where scene **120** is empty (without any targets present) and with transmitting antenna(s) **114** and receiving antenna(s) **116** located at 1 meter from the floor.

(34) As shown in FIG. **3**, the first range bins (bin **27** and lower) of the range spectrum **300** are dominated by the TX-RX crosstalk. The low frequency components of the TX-RX crosstalk are caused by the direct transmission from the transmit path **113** to the receive path **115**, which leads to a very small round-trip delay and thus a very small beat frequency of the beat signal $x_{\text{sub.IF}}(t)$. The reflections that are present after range bin **27** are due to reflections from the ground, e.g. at approximately 1 meter distance due to the 1 meter height of the radar above the ground.

(35) As illustrated in FIG. **3**, the TX-RX crosstalk may be a dominant low-frequency component in the receive signal, which may be modelled as target with a spread function $g_{\text{sub.0}}(T)$ with a bounded support on $[0, \tau_{\text{sub.L}}]$ with $\tau_{\text{sub.L}} \ll 1$ as the crosstalk covers the lower part of the spectrum only, where L correspond to the bin representing the highest bin associated with the TX-RX crosstalk (bin **27** in the example illustrated in FIG. **3**), τ denoting the normalized beat frequency, where $\tau \in (0,1]$, and where $\tau=1$ corresponds to the maximum unambiguous range.

(36) As illustrated in FIG. **3**, the TX-RX crosstalk may cause the detection of static objects at close ranges to be extremely challenging. This is particularly the case if the targets have a comparably small radar cross-section (RCS). For example, FIG. **4** shows exemplary range spectrum **400** associated with millimeter-wave radar sensor **102**, according to an embodiment of the present invention. Range spectrum **400** is obtained in a similar manner as range spectrum **300**, except that scene **120** includes a plastic pole instead of an empty scene. As shown in FIG. **4**, detecting the plastic pole target, which is located at a distance that corresponds to bin **17**, may be difficult in the presence of the TX-RX crosstalk.

(37) FIG. **5** shows waveforms **502** and **504** of radar data $x_{\text{sub.out_dig}}(n)$ associated with an empty scene, and with a scene with a target at 1 meter, respectively, according to an embodiment of the present invention.

(38) As illustrated by curve **502** (and as also illustrated in FIG. **3**), the contribution of the TX-RX crosstalk has a low frequency. As illustrated by curve **504** (and as also illustrated in FIG. **4**), when a target at close-range is present (e.g., in the field-of-view of millimeter-wave radar sensor **102** and close to the millimeter-wave radar sensor, such as within 1 meter), the reflected signals (**108**) are superimposed with the TX-RX crosstalk.

(39) As illustrated in FIGS. 3-5, it may be desirable to compensate for TX-RX crosstalk, e.g., before performing other radar processing tasks, such as target detection, tracking, and/or classification. FIG. 6 shows a flow chart of embodiment method 600 for dynamic TX-RX crosstalk cancellation, according to an embodiment of the present invention. Method 600 may be implemented by processing system 104.

(40) During step 602, a millimeter-wave radar sensor (e.g., 102), transmits radar signals (e.g., 106) organized in frames using one or more transmitting antennas (e.g., 114).

(41) During step 604, the millimeter-wave radar sensor receives signals using one or more receiving antennas (e.g., 116), which may include reflected radar signals (e.g., 108). The received signals may be organized in frames corresponding to the frames transmitted during step 602.

(42) During step 606, raw digital data $x_{\text{sub.out_dig}}(n)$ is generated based on the received signals (which includes TX-RX crosstalk and may include reflected radar signals 108). For example, in some embodiments, during step 606, the transmitted and received signals are mixed (e.g., using mixer 146) to generate an IF signal $x_{\text{sub.IF}}(t)$. The IF signal $x_{\text{sub.IF}}(t)$ is then filtered (e.g., with a low-pass and/or band-pass filter) and digitized with an ADC (e.g., 112) to generate the radar data $x_{\text{sub.out_dig}}(n)$.

(43) During step 610, a set $X_{\text{sub.S}}$ of the radar data $x_{\text{sub.out_dig}}(n)$ is selected. For example, in some embodiments, the set $X_{\text{sub.S}}$ of radar data $x_{\text{sub.out_dig}}(n)$ corresponds to a selection of a custom character of chirps with $|X_{\text{sub.S}}| = S \in \{1, \dots, N_{\text{sub.CN.sub.R}}\}$ from radar data $x_{\text{sub.out_dig}}(n)$, where $X_{\text{sub.S}} \in \text{custom character}^{\text{sup.N.sub.S.sub.} \times S}$ denote the chirps corresponding to the selection of a custom character arranged in a matrix. In some embodiments, the set $X_{\text{sub.S}}$ includes all radar data $x_{\text{sub.out_dig}}(n)$ (the selection of a custom character of chirps includes all chirps in the radar data $x_{\text{sub.out_dig}}(n)$). In some embodiments, the set $X_{\text{sub.S}}$ includes some, but not all, radar data $x_{\text{sub.out_dig}}(n)$.

(44) During step 612, a crosstalk approximation of the TX-RX crosstalk is determined, e.g., by using a least-squares fit of a low-order polynomial performed on the set $X_{\text{sub.S}}$ of radar data $x_{\text{sub.out_dig}}(n)$. As illustrated by curve 502, in some embodiments, such least-squares fit may represent a good approximation of the TX-RX crosstalk. In some embodiments, a polynomial of order lower than 10, such as 6, may be used.

(45) For example, in some embodiments, the least-squares fit is performed for each column of $X_{\text{sub.S}}$. For example, without loss of generality, letting the sampling times be $n=1, \dots, N_{\text{sub.S}}$, and considering a polynomial of degree d given as

$$p_{\text{sub.d}}(n;a) = \sum_{k=0}^d a(k) n^{\text{sup.k}}, \quad (1)$$

where $a=[a(0), \dots, a(d)] \in \text{custom character}^{\text{sup.d}}$ with $d=d+1$ and $a \in \text{custom character}^{\text{sup.d}}$ parameterizes the polynomial, where

$$p_{\text{sub.d}}(a) = [p_{\text{sub.d}}(1;a), \dots, p_{\text{sub.d}}(N_{\text{sub.S}};a)] \in \text{custom character}^{\text{sup.N.sub.S}} \quad (2)$$

and

$$P_{\text{sub.d}}(A) = [p_{\text{sub.d}}(a_{\text{sub.1}}), \dots, p_{\text{sub.d}}(a_{\text{sub.S}})] \in \text{custom character}^{\text{sup.N.sub.S.sub.} \times S} \quad (3)$$

with

$$A = [a_{\text{sub.1}}, \dots, a_{\text{sub.S}}] \in \text{custom character}^{\text{sup.d} \times S} \quad (4)$$

the least-square problem may be given as

$$(46) \min_A \|\mathbf{X}_S - P_d(A)\|_F, \quad (5)$$

where $\|\cdot\|_F$ represents the Frobenius norm. Letting the matrix $B \in \text{custom character}^{\text{sup.N.sub.S.sub.} \times d}$ be a Vandermonde matrix with elements $B(i,j)=i^{\text{sup.j-1}}$ such that $p_{\text{sub.d}}(a)=Ba$ and $P_{\text{sub.d}}(A)=BA$, the linear least-squares problem may be given as

$$(47) \min_A \|\mathbf{X}_S - BA\|_F, \quad (6)$$

which as $B \in \text{custom character}^{\text{sup.N.sub.S.sub.} \times d}$ has full rank has a unique optimizer if d

N.sub.s given as

$$A.\text{sub.opt}=(B'B).\text{sup.}-1B'X.\text{sub.S} \quad (7)$$

and a unique optimal value, where B' refers to the transpose of B if B is a real matrix and to the conjugate transpose (Hermitian) of B if B is a complex matrix. The fitted polynomials may be given as

$$P.\text{sub.d}(A.\text{sub.opt})=B(B'B).\text{sup.}-1B'X.\text{sub.S}=P.\text{sub.BX}.\text{sub.S} \quad (8)$$

and may be obtained by the orthogonal projection $P.\text{sub.BX}.\text{sub.S}$ of $X.\text{sub.S}$ onto $\text{range}(B)$.

(48) During step 614, the computation of a residual of the estimation of the TX-RX crosstalk is performed (the computation of the residual of the crosstalk approximation performed during step 612). For example, in some embodiments, the residual of the polynomial fit (computed during step 612) is computed, e.g., by subtracting the polynomial fit from the set $X.\text{sub.S}$. For example, in some embodiments, the residual $X.\text{sub.R}$ may be computed as

$$X.\text{sub.R}=X.\text{sub.S}-P.\text{sub.d}(A.\text{sub.opt})=P.\text{sub.B}.\text{sup.}\perp X.\text{sub.S}, \quad (9)$$

which corresponds to the projection of $X.\text{sub.S}$ onto $(\text{range}(B)).\text{sup.}\perp$, where $B'B$ as well as $B'X.\text{sub.S}$ can be obtained in closed form.

(49) As illustrated in FIG. 5, the residual $X.\text{sub.R}$ may be indicative of the presence of close-range target. During step 616, a spectrum is determined based on residual $X.\text{sub.R}$ using a spectral estimator. For example, in some embodiments, considering real-valued raw ADC data $X.\text{sub.S}$ (although the approach may not be limited to real-valued raw ADC data and may be applied, e.g., to real and imaginary part of complex-valued ADC data), the range spectrum of $X.\text{sub.S}$ is symmetric by construction and only one half of the range spectrum contains useful information. Thus, the spectral estimator may be defined as a function mapping from  $\text{sup.N}.\text{sup.S}.\text{sup.}\times S$ to

$$(50) \mathbb{C}^{\frac{q}{2}N_S \times S},$$

only, where the additional factor q accounts for spectral estimators which estimate the spectrum on, e.g., an oversampled FFT grid. For example, in some embodiments, the spectral estimator

 $\text{custom character.sub.R}$ (where

$$(51) \mathcal{F}_R: \mathbb{R}^{N_S \times S} \xrightarrow{\text{fwdarw.}} \mathbb{C}^{\frac{q}{2}N_S \times S}, X_R \mapsto \mathcal{F}_R(X_R))$$

estimates the useful part of the range spectrum of $X.\text{sub.R}$ using a spectral grid with

$$(52) \frac{q}{2}N_S$$

grid points and may generate the range spectrum using an FFT (with time-domain windowing) along the first dimension of $X.\text{sub.R}$ and by considering the first half of the FFT spectrum only. In some embodiments, the spectra PR of the residual $X.\text{sub.R}$ may be given by

$$P.\text{sub.R}=\text{custom character.sub.R}(X.\text{sub.R}) \quad (10)$$

(53) During step 618, the first L bins of the spectra $P.\text{sub.R}$ of the residual $X.\text{sub.R}$ are integrated. For example, in some embodiments, the first L bins of the range spectra $P.\text{sub.R}$ of the residuum along all S is incoherently (non-coherently) integrated to obtain the spectrum $P.\text{sub.D} \in$

 $\text{custom character.sup.L}$, which may be computed as

$$P.\text{sub.D}=(P.\text{sub.R}(1:L,:)\odot P^*.\text{sub.R}(1:L,:))1.\text{sub.S}\times 1 \in \text{custom character.sup.L}, \quad (11)$$

where $\odot$ denotes the element-wise Hadamard matrix product, $P.\text{sub.R}^*$ denotes the complex conjugate of $P.\text{sub.R}$, and $1.\text{sub.S}\times 1$ denotes the all-ones vector of dimension S, and where

$$(54) L(0 < L \leq \frac{q}{2}N_S)$$

corresponds to the last range bin that is significantly affected by the TX-RX crosstalk.

(55) In some embodiments, spectrum $P.\text{sub.D}$ is indicative of whether a close-range target is present. During step 620, target detection is performed, e.g., using constant false alarm rate (CFAR). In some embodiments, the CFAR may be aborted after the first target is detected.

(56) If no target is detected during step 622, a crosstalk compensation factor $P.\text{sub.est}$ is updated during step 624 (where the crosstalk compensation factor $P.\text{sub.est}$ may be previously initialized, e.g., to a default value). For example, let $P.\text{sub.est} \in \text{custom character}$

.sup.L×N.sup.C.sup.N.sup.R denote the current estimate of the TX-RX crosstalk spectrum (e.g., which is stored in a memory of processing system **104**). In addition, let

$$(57) \mathcal{F}: \mathbb{R}^{N_S \times N_C \times N_R} \rightarrow \mathbb{C}^{N_S \times N_C \times N_R}, X \mapsto \mathcal{F}(X)$$

denote a spectral estimator which estimates the range spectra of every chirp and receive channel in the radar data X of the current frame, the estimate of the spectrum of the TX-RX crosstalk may be updated according to an exponential moving average (EMA) given as

$$P_{\text{sub.est}} \leftarrow \alpha P_{\text{sub.est}}(1-\alpha)P(1:L, :, :), \quad (12)$$

where $P(1:L, :, :)$ denotes the first L range bins of the estimated range spectra of the raw ADC data $x_{\text{sub.out_dig}}(n)$ (of $P = \text{custom_character}(X)$, where X represents the radar data $x_{\text{sub.out_dig}}(n)$) and α with $0 \leq \alpha \leq 1$ denotes the exponential weighting factor of the EMA, and where $\leftarrow$ is a symbol that implies that the operation to the right of the symbol $\leftarrow$ is performed first and then assigned to the variable to the left of the symbol $\leftarrow$. The updated crosstalk compensation factor $P_{\text{sub.est}}$ may be stored (e.g., replacing the previous crosstalk compensation factor $P_{\text{sub.est}}$) in (e.g., volatile or non-volatile) memory (e.g., of processing system **104**).

(58) If a target is detected during step **622**, step **624** is skipped.

(59) During step **626** (which may be performed before, during, or after steps **608** and **624** (if applicable)), the spectrum of the radar data is estimated. For example, in some embodiments, the spectrum P of the radar data $x_{\text{sub.out_dig}}(n)$ may be given as

$$P = \text{custom_character}(X) \quad (13)$$

where $\text{custom_character}()$ is the same spectral estimator used during step **616** and where X represents the radar data $x_{\text{sub.out_dig}}(n)$. In some embodiments, the spectral estimator $\text{custom_character}()$ is a range FFT. In some embodiments, the spectral estimator $\text{custom_character}()$ is a Capon estimator. Other estimators may also be used for performing steps **626** and **616**.

(60) During step **628**, the spectrum generated during step **626** is compensated based on the stored crosstalk compensation factor $P_{\text{sub.est}}$. For example, in some embodiments, the spectrum P generated during step **626** is compensated as

$$P(1:L, :, :) \leftarrow P(1:L, :, :) - P_{\text{sub.est}} \quad (14)$$

In some embodiments, L may be, e.g., $qN_{\text{sub.S}}/2$. In some embodiments, the compensated spectrum generated during step **626** is advantageously not affected (or not substantially affected) by TX-RX crosstalk.

(61) During step **630**, the compensated spectrum P is used, e.g., to detect, track and/or classify targets in ways known in the art.

(62) As shown in FIG. **6**, step **608** (which includes **610**, **612**, **614**, **616**, **618**, **620**, and **622**) operates as a classifier that classifies $X_{\text{sub.S}}$ as including TX-RX crosstalk or including TX-RX crosstalk plus target(s). As also illustrated in FIG. **6**, the crosstalk compensation factor $P_{\text{sub.est}}$ (which is indicative of the TX-RX crosstalk between transmit path **113** and receive path **115**) is updated (in step **624**) only when no target is detected in the chirps of set $X_{\text{sub.S}}$. By only updating the crosstalk compensation factor $P_{\text{sub.est}}$ when no target is detected, some embodiments advantageously compensate for TX-RX crosstalk without including contributions from close-range target(s), increasing the performance of the crosstalk cancellation.

(63) As shown in FIG. **6**, in some embodiments, step **618** may be performed to generate an integrated spectrum of the residual by performing incoherent integration. In some embodiments, coherent integration may be performed instead (e.g., with an additional beamforming step). In some embodiments, step **618** may be omitted and step **620** may be performed in a non-integrated spectrum of the residual (e.g., generated during step **618**).

(64) In some embodiments, step **608** and step **624** (if applicable) are performed with the same rate as the frame rate of the radar data. By performing TX-RX crosstalk cancellation at the same rate as the frame rate, some embodiments advantageously adapt to new environments quickly. For example, in some embodiments, millimeter-wave radar sensor **102** may be implemented in a

mobile device (such as a smartphone) and may move quickly, such as by rotating or displacing. A fast TX-RX crosstalk cancellation rate may advantageously allow for close-range target detection, which may improve the performance of, e.g., hand gesture recognition for operating the smartphone.

(65) In some embodiments, step **608** and step **624** (if applicable) are performed with rate slower than the frame rate of the radar data (such as 10 times slower or 100 times slower). By performing TX-RX crosstalk cancellation at a rate slower than the frame rate, some embodiments advantageously save computational power. For example, in some embodiments, the environment influencing the TX-RX crosstalk may change at a lower rate. For example, a millimeter-wave radar sensor that is not mobile and is placed in an indoor environment (such as at the ceiling of a room) may not experience fast changes in the TX-RX crosstalk (for example, changes due to temperature may have a slow time constant). Thus, some embodiments may advantageously save computational power by using a slow rate of TX-RX crosstalk estimation without substantially impacting the TX-RX crosstalk cancellation performance. In some embodiments, the TX-RX crosstalk cancellation (e.g., step **628**) may be performed at a faster rate (e.g., every frame) than the rate of performing the TX-RX crosstalk estimation (e.g., step **608** and **624** (if applicable)).

(66) In some embodiments, the TX-RX crosstalk cancellation rate may be dynamically changed. For example, in an embodiment in which millimeter-wave radar sensor **102** is implemented in a vehicle, the TX-RX crosstalk cancellation rate may be based on the speed of the vehicle. For example, when the vehicle is stopped at a red light, the TX-RX crosstalk cancellation rate may be slow (e.g., 10 times or 100 times slower than the radar frame rate). Having a slow TX-RX crosstalk cancellation rate when the vehicle is stopped may advantageously save computational power and battery power without substantially impacting the TX-RX crosstalk cancellation performance (since the environment may not change very rapidly). When the vehicle is moving at a high speed in a highway, the TX-RX crosstalk cancellation rate may be fast (e.g., at the same rate as the frame rate of the radar), which may advantageously improve the performance of the TX-RX crosstalk cancellation as the vehicle moves and turns at high speed).

(67) Advantages of some applications include improved target detection performance for close-range targets, e.g., when compared with radars that use HPFs to address crosstalk (which may introduce a minimum range constraint). Additional advantages include the ability to cancel TX-RX crosstalk in applications for detecting static targets.

(68) As shown in FIG. 6, the TX-RX crosstalk compensation may be performed in the frequency domain (as illustrated by step **628**). In some embodiments, the TX-RX crosstalk compensation may be performed in the time domain. For example, FIG. 7, shows a flow chart of embodiment method **700** for dynamic TX-RX crosstalk cancellation, according to an embodiment of the present invention. Method **700** may be implemented by processing system **104**. Method **700** is similar and may be performed in a similar manner as method **600**. Method **700**, however, includes steps **724** and **728** for compensating for TX-RX crosstalk in the time domain, and step **626** is performed on the compensated time-domain radar data $x_{\text{sub.out_dig_comp}}(n)$ instead of on $x_{\text{sub.out_dig}}(n)$.

(69) For example, during step **724** (when no target is detected in the set $X_{\text{sub.S}}$ of the radar data $x_{\text{sub.out_dig}}(n)$), a crosstalk compensation factor $X_{\text{sub.crosstalk}}$ is updated, where $X_{\text{sub.crosstalk}}$ is a time domain compensation factor ($X_{\text{sub.crosstalk}} \in \text{custom character}^{\text{sup.N.sup.S.sup.} \times \text{sup.N.sup.C.sup.} \times \text{sup.N.sup.R}}$) and where the compensation factor $X_{\text{sub.crosstalk}}$ may be previously initialized, e.g., to a default value. For example, in some embodiments, $X_{\text{sub.crosstalk}}$ corresponds to the best fit curve generated during step **612**, and which may substantially approximate the TX-RX crosstalk. Updating the crosstalk compensation factor $X_{\text{sub.crosstalk}}$ may be performed using EMA, e.g., as

$$X_{\text{sub.crosstalk}} \leftarrow \alpha X_{\text{sub.crosstalk}} + (1 - \alpha) \text{custom character}(P_{\text{sub.d}}(A_{\text{sub.opt}})), \quad (15)$$

where $\text{custom character}^{\text{sup.N.sup.S.sup.} \times \text{sup.N.sup.C.sup.} \times \text{sup.N.sup.R}}$, $P_{\text{sub.d}}(A_{\text{sub.opt}}) \in \text{custom character}^{\text{sup.N.sup.S.sup.} \times \text{sup.N.sup.C.sup.} \times \text{sup.N.sup.R}}$, $P_{\text{sub.d}}(A_{\text{sub.opt}})$ is a function of $A_{\text{sub.opt}}$ and α is a scalar value.

(P.sub.d(A.sub.opt)) maps the least-squares fit P.sub.d(A.sub.opt) based on the selection of a custom character to the corresponding elements in X.sub.crosstalk. In some embodiments, the selection of a custom character contains at least one chirp of each receive channel to obtain a reasonable mapping of a custom character.

(70) In some embodiments, the compensated radar data (comp) may be given as

$$X_{\text{sub.comp}} = X - X_{\text{sub.crosstalk}}, \quad (16)$$

where X represents the radar data $x_{\text{sub.out_dig}}(n)$.

(71) During step 730, the spectrum of the compensated radar data (comp) is performed, e.g., in a similar manner as in step 730. For example, in some embodiments, the spectrum of the compensated radar data is performed as

$$P = \text{custom_character}(X_{\text{sub.comp}}) \quad (17)$$

where custom_character() is the same spectral estimator used during step 616.

(72) Advantages of some embodiment implementing the TX-RX cancellation in the time domain (e.g., method 700) include achieving TX-RX cancellation without suffering from residual side lobes of the TX-RX crosstalk.

(73) Example embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

(74) Example 1. A method including: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

(75) Example 2. The method of example 1, further including: determining a crosstalk approximation of the transmitter-receiver crosstalk on the set of data; determining a residual of the crosstalk approximation based on the polynomial fit and the set of data; and determining a spectrum of the residual of the crosstalk approximation, where performing the target detection on the set of data includes performing the target detection on the set of data based on the spectrum of the residual of the polynomial fit.

(76) Example 3. The method of examples 1 or 2, further including: determining a polynomial fit on the set of data; determining a residual of the polynomial fit based on the polynomial fit and the set of data; and determining a spectrum of the residual of the polynomial fit, where performing the target detection on the set of data includes performing the target detection on the set of data based on the spectrum of the residual of the polynomial fit.

(77) Example 4. The method of one of examples 1 to 3, where determining the polynomial fit includes performing a least-squares fit of low-order polynomial on the set of data, where the low-order polynomial is a polynomial of order between 6 and 10.

(78) Example 5. The method of one of examples 1 to 4, further including integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, where performing the target detection on the set of data includes performing the target detection on the set of data based on the integrated spectrum.

(79) Example 6. The method of one of examples 1 to 5, where integrating the first L bins of the spectrum of the residual of the polynomial fit includes incoherently integrating the first L bins of the spectrum of the residual of the polynomial fit.

(80) Example 7. The method of one of examples 1 to 6, where generating the radar spectrum and generating the spectrum of the residual of the polynomial fit includes using the same spectral

estimator.

(81) Example 8. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a same rate as a frame rate of the radar data.

(82) Example 9. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a rate lower than a frame rate of the radar data.

(83) Example 10. The method of one of examples 1 to 7, where selecting the set of data and performing the target detection on the set of data includes selecting the set of data and performing the target detection on the set of data at a dynamic rate.

(84) Example 11. The method of one of examples 1 to 10, further including generating the radar spectrum based on the radar data and the crosstalk compensation factor at a faster rate than performing the target detection on the set of data.

(85) Example 12. The method of one of examples 1 to 11, where performing the target detection on the set of data includes performing the target detection on the set of data using a constant false alarm rate (CFAR).

(86) Example 13. The method of one of examples 1 to 12, further including aborting the target detection on the set of data when a first target is detected.

(87) Example 14. The method of one of examples 1 to 13, where updating the crosstalk compensation factor includes updating the crosstalk compensation factor using an exponential moving average.

(88) Example 15. The method of one of examples 1 to 14, where the crosstalk compensation factor includes a crosstalk spectrum, and where generating the radar spectrum includes: generating a first radar spectrum based on the radar data; and subtracting the crosstalk spectrum from the first radar spectrum.

(89) Example 16. The method of one of examples 1 to 14, where the crosstalk compensation factor includes a time-domain crosstalk estimate, and where generating the radar spectrum includes: subtracting the time-domain crosstalk estimate from the radar data to generate a compensated radar data; and generating the radar spectrum based on the compensated radar data.

(90) Example 17. The method of one of examples 1 to 16, where the set of data includes a plurality of linear chirps.

(91) Example 18. The method of one of examples 1 to 17, further including generating the radar data based on signals from the receive path of the radar sensor.

(92) Example 19. The method of one of examples 1 to 18, further including performing target detection or target tracking based on the radar spectrum.

(93) Example 20. The method of one of examples 1 to 19, where the radar sensor is a millimeter-wave radar sensor.

(94) Example 21. The method of one of examples 1 to 20, where the set of data is a sub-set of radar data selected from the radar data.

(95) Example 22. A radar system including: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and a processing system configured to: initialize a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, select a set of data from the radar data, perform target detection on the set of data, and after performing the target detection on the set of data, when no target is detected in the set of data, update the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generate a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generate the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor.

(96) Example 23. A method including: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the a transmit path of a millimeter-wave radar sensor and a receive path of the millimeter-wave radar sensor; receiving radar data from the millimeter-wave radar sensor; selecting a sub-set of data from the radar data; determining a polynomial fit on the sub-set of data; determining a residual of the polynomial fit based on the polynomial fit and the sub-set of data; and determining a spectrum of the residual of the polynomial fit; incoherently integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than a maximum number of bins of the spectrum of the residual of the polynomial fit; performing target detection on the sub-set of data based on the integrated spectrum; after performing the target detection on the sub-set of data, when no target is detected in the sub-set of data, updating the crosstalk compensation factor based on the sub-set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the sub-set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; and performing target detection or target tracking based on the radar spectrum.

(97) Example 24. A method including: receiving radar data from a radar sensor; selecting a set of data from the radar data; determining a crosstalk approximation of a transmitter-receiver crosstalk between a transmit path of the radar sensor and a receive path of the radar sensor based on the set of data; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; determining a spectrum of the residual of the crosstalk approximation; and performing target detection on the set of data based on the spectrum of the residual of the crosstalk approximation.

(98) Example 25. The method of example 24, where determining the crosstalk approximation includes determining a polynomial fit on the set of data, and where determining the residual of the crosstalk approximation includes subtracting the crosstalk approximation from the set of data.

(99) While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method comprising: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; determining a crosstalk approximation of the transmitter-receiver crosstalk; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; and determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, and the spectrum of the residual of the crosstalk approximation comprises range bins representing frequency domain values.

2. The method of claim 1, wherein determining the crosstalk approximation comprises determining

a polynomial fit on the set of data, wherein determining the residual of the crosstalk approximation comprises determining a residual of the polynomial fit based on the polynomial fit and the set of data, and wherein determining the spectrum of the residual of the crosstalk approximation comprises determining a spectrum of the residual of the polynomial fit.

3. The method of claim 2, wherein determining the polynomial fit comprises performing a least-squares fit of a low-order polynomial on the set of data, wherein the low-order polynomial is a polynomial of order between 6 and 10.

4. The method of claim 2, further comprising integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the integrated spectrum.

5. The method of claim 4, wherein integrating the first L bins of the spectrum of the residual of the polynomial fit comprises incoherently integrating the first L bins of the spectrum of the residual of the polynomial fit.

6. The method of claim 1, wherein: generating the radar spectrum comprises using a spectral estimator; and generating the spectrum of the residual of the crosstalk approximation comprises using the spectral estimator.

7. The method of claim 1, wherein performing the target detection on the set of data comprises performing the target detection on the set of data using a constant false alarm rate (CFAR).

8. The method of claim 1, further comprising aborting the target detection on the set of data when a first target is detected.

9. The method of claim 1, wherein updating the crosstalk compensation factor comprises updating the crosstalk compensation factor using an exponential moving average.

10. The method of claim 1, wherein the crosstalk compensation factor comprises a crosstalk spectrum, and wherein generating the radar spectrum comprises: generating a first radar spectrum based on the radar data; and subtracting the crosstalk spectrum from the first radar spectrum.

11. The method of claim 1, wherein the crosstalk compensation factor comprises a time-domain crosstalk estimate, and wherein generating the radar spectrum comprises: subtracting the time-domain crosstalk estimate from the radar data to generate compensated radar data; and generating the radar spectrum based on the compensated radar data.

12. The method of claim 1, wherein the set of data comprises a plurality of linear chirps.

13. The method of claim 1, further comprising generating the radar data based on signals from the receive path of the radar sensor.

14. The method of claim 1, further comprising performing the target detection or target tracking based on the radar spectrum.

15. The method of claim 1, wherein the radar sensor is a millimeter-wave radar sensor.

16. The method of claim 1, wherein the set of data is a subset of radar data selected from the radar data.

17. The method of claim 1, wherein: selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a same rate as a frame rate of the radar data; or selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a rate lower than the frame rate of the radar data; or selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a dynamic rate; or the method further comprises generating the radar spectrum based on the radar data and the crosstalk compensation factor at a faster rate than performing the target detection on the set of data.

18. A radar system comprising: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of

the receive path; and at least one processor and at least one memory with instructions stored thereon, wherein the instructions, when executed by the at least one processor enable the radar system to perform the steps of: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, selecting a set of data from the radar data, performing target detection on the set of data, and after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; determining a crosstalk approximation of the transmitter-receiver crosstalk; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; and determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, and the spectrum of the residual of the crosstalk approximation comprises range bins representing frequency domain values.

19. The radar system of claim 18, wherein: selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a same rate as a frame rate of the radar data, or selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a rate lower than the frame rate of the radar data, or selecting the set of data and performing the target detection on the set of data comprises selecting the set of data and performing the target detection on the set of data at a dynamic rate, or the instructions, when executed by the at least one processor further enable the radar system to perform the step of generating the radar spectrum based on the radar data and the crosstalk compensation factor at a faster rate than performing the target detection on the set of data.

20. A method comprising: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between a transmit path of a radar sensor and a receive path of the radar sensor; receiving radar data from the radar sensor; selecting a set of data from the radar data; performing target detection on the set of data; after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor; determining a crosstalk approximation of the transmitter-receiver crosstalk; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data; and determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, wherein determining the crosstalk approximation comprises determining a polynomial fit on the set of data, wherein determining the residual of the crosstalk approximation comprises determining a residual of the polynomial fit based on the polynomial fit and the set of data, and wherein determining the spectrum of the residual of the crosstalk approximation comprises determining a spectrum of the residual of the polynomial fit.

21. The method of claim 20, wherein determining the polynomial fit comprises performing a least-squares fit of a low-order polynomial on the set of data, wherein the low-order polynomial is a polynomial of order between 6 and 10.

22. The method of claim 20, further comprising integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the

polynomial fit, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the integrated spectrum.

23. The method of claim 22, wherein integrating the first L bins of the spectrum of the residual of the polynomial fit comprises incoherently integrating the first L bins of the spectrum of the residual of the polynomial fit.

24. A radar system comprising: a radar sensor configured to transmit radar signals via a transmit path, receive reflected radar signals via a receive path, and provide radar data based on an output of the receive path; and at least one processor and at least one memory with instructions stored thereon, wherein the instructions, when executed by the at least one processor enable the radar system to perform the steps of: initializing a crosstalk compensation factor indicative of a transmitter-receiver crosstalk between the transmit path and the receive path, selecting a set of data from the radar data, performing target detection on the set of data, after performing the target detection on the set of data, when no target is detected in the set of data, updating the crosstalk compensation factor based on the set of data and, after updating the crosstalk compensation factor, generating a radar spectrum based on the radar data and the crosstalk compensation factor, and when a target is detected in the set of data, generating the radar spectrum based on the radar data and the crosstalk compensation factor without updating the crosstalk compensation factor, determining a crosstalk approximation of the transmitter-receiver crosstalk; determining a residual of the crosstalk approximation based on the crosstalk approximation and the set of data, and determining a spectrum of the residual of the crosstalk approximation, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the spectrum of the residual of the crosstalk approximation, wherein determining the crosstalk approximation comprises determining a polynomial fit on the set of data, wherein determining the residual of the crosstalk approximation comprises determining a residual of the polynomial fit based on the polynomial fit and the set of data, and wherein determining the spectrum of the residual of the crosstalk approximation comprises determining a spectrum of the residual of the polynomial fit.

25. The radar system of claim 24, wherein determining the polynomial fit comprises performing a least-squares fit of a low-order polynomial on the set of data, wherein the low-order polynomial is a polynomial of order between 6 and 10.

26. The radar system of claim 24, wherein the instructions, when executed by the at least one processor further enable the radar system to perform the step of integrating a first L bins of the spectrum of the residual of the polynomial fit to generate an integrated spectrum, L being a positive integer greater than 1 and lower than or equal to a maximum number of bins of the spectrum of the residual of the polynomial fit, wherein performing the target detection on the set of data comprises performing the target detection on the set of data based on the integrated spectrum.
